Section 12.4: The Cross Product

Vectors and the Geometry of Space

The Cross Product — A New Kind of Product

The **dot product** takes two vectors and returns a **scalar** ($\vec{a} \cdot \vec{b} \in \mathbb{R}$).
Now we introduce a second way to multiply vectors:

> The **cross product** $\vec{a} \times \vec{b}$ takes two vectors in $\mathbb{R}^3$ and returns a **vector** that is **perpendicular** to both $\vec{a}$ and $\vec{b}$.

- The cross product is **only defined in 3D** (not 2D!)
- It's used everywhere: torque, angular momentum, normal vectors to surfaces, area calculations

Quick Review: Determinants

2 $\times$ 2 Determinant

$$|a \quad b\; c \quad d| = ad - bc$$

Example 1

Compute the determinant $|2 \quad 3\; 4 \quad 1|$.

$$|2 \quad 3\; 4 \quad 1| = (2)(1) - (3)(4) = 2 - 12 = -10$$

3 $\times$ 3 Determinant (Cofactor Expansion)

Expand along the **first row**:

$$\left|\; a_1 \quad a_2 \quad a_3\; b_1 \quad b_2 \quad b_3\; c_1 \quad c_2 \quad c_3 \;\right| = a_1 \left| b_2 \quad b_3\; c_2 \quad c_3 \right| - a_2 \left| b_1 \quad b_3\; c_1 \quad c_3 \right| + a_3 \left| b_1 \quad b_2\; c_1 \quad c_2 \right|$$

Notice the **alternating signs**: $+$, $-$, $+$.

Example 2

Compute the determinant:

$$|2 \quad 3 \quad 1\; 4 \quad 1 \quad -3\; -2 \quad 5 \quad 2|$$

Expand along the first row using the cofactor formula.

Setup: Expand along the first row with alternating signs $+$, $-$, $+$:

$$= 2\,|1 \quad -3\; 5 \quad 2| - 3\,|4 \quad -3\; -2 \quad 2| + 1\,|4 \quad 1\; -2 \quad 5|$$

Expand each 2x2 determinant:

$$= 2(1 \cdot 2 - (-3) \cdot 5) - 3(4 \cdot 2 - (-3)(-2)) + 1(4 \cdot 5 - 1 \cdot (-2))$$

Simplify:

$$= 2(2 + 15) - 3(8 - 6) + 1(20 + 2) = 2(17) - 3(2) + 1(22) = 34 - 6 + 22 = 50$$

Note: this gives a **scalar**.

The Cross Product Formula

Cross Product

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then:

$$\vec{a} \times \vec{b} = \left| \vec{i} \quad \vec{j} \quad \vec{k} \, a_1 \quad a_2 \quad a_3 \, b_1 \quad b_2 \quad b_3 \right|$$

Expanding along the first row:

$$\vec{a} \times \vec{b} = \vec{i}(a_2b_3 - a_3b_2) - \vec{j}(a_1b_3 - a_3b_1) + \vec{k}(a_1b_2 - a_2b_1)$$

The result is a **vector** (note the $\vec{i}$, $\vec{j}$, $\vec{k}$). This is a "symbolic" determinant — the first row contains vectors, not numbers.

Example 3: Computing a Cross Product

Example 3

Let $\vec{a} = \langle -1, 3, 4 \rangle$ and $\vec{b} = \langle 5, 2, 1 \rangle$. Find $\vec{a} \times \vec{b}$.

Set up the 3x3 determinant and expand along the first row.

Step 1 — Set up the determinant:

$$\vec{a} \times \vec{b} = \left| \vec{i} \quad \vec{j} \quad \vec{k} \, -1 \quad 3 \quad 4 \, 5 \quad 2 \quad 1 \right|$$

Step 2 — Expand:

$$= \vec{i}(3 \cdot 1 - 4 \cdot 2) - \vec{j}((-1) \cdot 1 - 4 \cdot 5) + \vec{k}((-1) \cdot 2 - 3 \cdot 5)$$

Step 3 — Simplify:

$$= \vec{i}(3 - 8) - \vec{j}(-1 - 20) + \vec{k}(-2 - 15)$$

$$= -5\,\vec{i} + 21\,\vec{j} - 17\,\vec{k} = \langle -5, 21, -17 \rangle$$

Verification: $\vec{a} \times \vec{b}$ should be perpendicular to both inputs, so the dot products should be zero:

$$\vec{a} \cdot (\vec{a} \times \vec{b}) = \langle -1, 3, 4 \rangle \cdot \langle -5, 21, -17 \rangle = 5 + 63 - 68 = 0 \checkmark$$

The result is a **vector**, and it's perpendicular to both inputs. Always verify with a quick dot product!

Your Turn: Cross Product Practice

Example 4

Compute $\langle 1, 0, 2\rangle \times \langle 3, 1, -1\rangle$.

Set up the determinant and expand. Check your answer with a dot product.

Determinant setup:

$$\left|\vec{i} \quad \vec{j} \quad \vec{k} 1 \quad 0 \quad 2 3 \quad 1 \quad -1\right|$$

Expand:

$$= \vec{i}(0 \cdot (-1) - 2 \cdot 1) - \vec{j}(1 \cdot (-1) - 2 \cdot 3) + \vec{k}(1 \cdot 1 - 0 \cdot 3)$$

$$= \vec{i}(0 - 2) - \vec{j}(-1 - 6) + \vec{k}(1 - 0)$$

$$= \langle -2, 7, 1\rangle$$

Check: $\langle 1, 0, 2\rangle \cdot \langle -2, 7, 1\rangle = -2 + 0 + 2 = 0 \checkmark$

Key Properties of the Cross Product

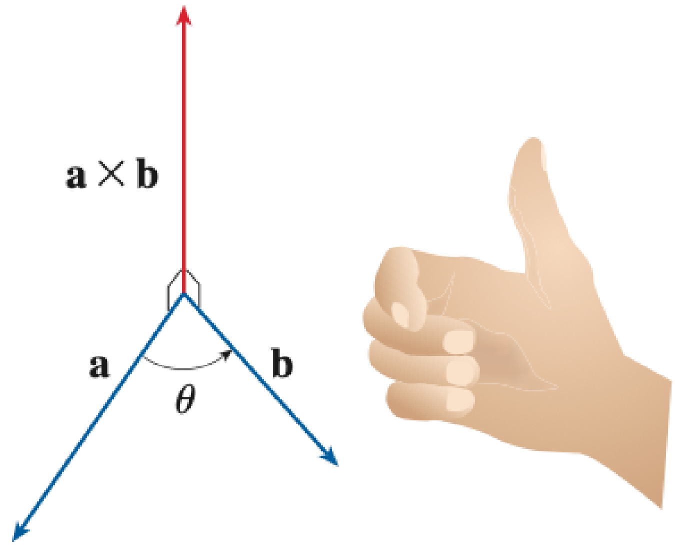
Here are the properties you'll use most often:

Property	Meaning
$\vec{a} \times \vec{a} = \vec{0}$	A vector is parallel to itself, so the cross product is the zero vector
$\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$	The result is perpendicular to both inputs
$\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$	**Anti-commutative** — swapping order flips the sign

Order matters! $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$ (they point in **opposite** directions).

Which direction does $\vec{a} \times \vec{b}$ point? Use the **right-hand rule**:

1. Point your fingers along $\vec{a}$
2. Curl them toward $\vec{b}$
3. Your **thumb** points in the direction of $\vec{a} \times \vec{b}$



Magnitude of the Cross Product

Magnitude Formula

$$|\vec{a} \times \vec{b}| = |\vec{a}|\,|\vec{b}|\,\sin\theta$$

where θ is the angle between $\vec{a}$ and $\vec{b}$ ($0 \leq \theta \leq \pi$).

Compare with the dot product:

- **Dot product:** $\vec{a} \cdot \vec{b} = |\vec{a}|\,|\vec{b}|\,\cos\theta$ — measures **alignment**
- **Cross product magnitude:** $|\vec{a} \times \vec{b}| = |\vec{a}|\,|\vec{b}|\,\sin\theta$ — measures **perpendicularity**

If $\theta = 0$ or $\theta = \pi$ (vectors are **parallel**), then $\sin\theta = 0$, so:

$\vec{a}$ and $\vec{b}$ are **parallel** if and only if $\vec{a} \times \vec{b} = \vec{0}$.

This gives us two tests:

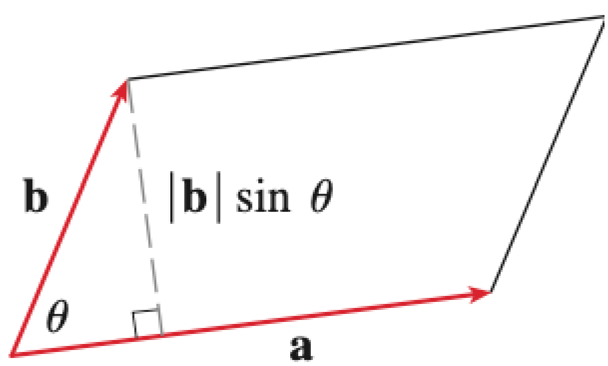
- **Parallel test:** $\vec{a} \times \vec{b} = \vec{0}$
- **Perpendicular test (from 12.3):** $\vec{a} \cdot \vec{b} = 0$

Area from the Cross Product

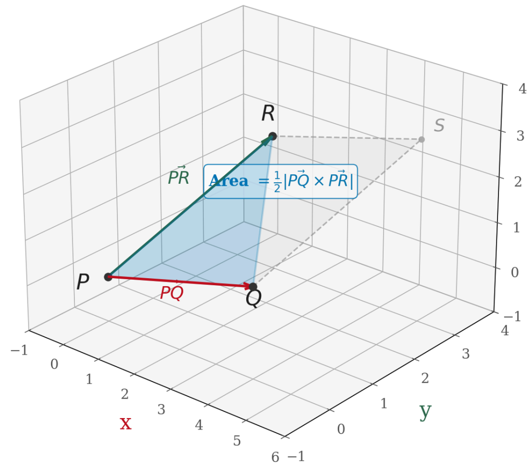
Recall that the area of a parallelogram is base $\times$ height:

$$\text{Area} = |\vec{a}|\,(|\vec{b}|\sin\theta) = |\vec{a} \times \vec{b}|$$

- **Parallelogram** with sides $\vec{a}$ and $\vec{b}$: Area $= |\vec{a} \times \vec{b}|$
- **Triangle** with sides $\vec{a}$ and $\vec{b}$: Area $= \frac{1}{2}|\vec{a} \times \vec{b}|$



This is incredibly useful: find the area of **any triangle in 3D** just from its three vertices!



R
S
P
Q
$\vec{PR}$
$\vec{PQ}$
Area $= \frac{1}{2}|\vec{PQ} \times \vec{PR}|$
x
y

Example 5: Normal Vector and Area from Three Points

Example 5

Given $P(-2,1,7)$, $Q(-3,4,11)$, and $R(3,3,8)$:

(a) Find a vector perpendicular to the plane through P, Q, R.

(b) Find the area of triangle PQR.

Form two edge vectors from P, then take their cross product.

Step 1 — Edge vectors:

$$\vec{PQ} = Q - P = \langle -3-(-2),\ 4-1,\ 11-7\rangle = \langle -1, 3, 4\rangle$$

$$\vec{PR} = R - P = \langle 3-(-2),\ 3-1,\ 8-7\rangle = \langle 5, 2, 1\rangle$$

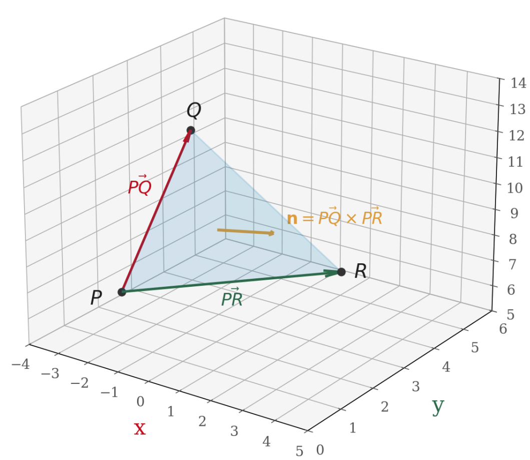
Step 2 — Normal vector (cross product — already computed in Example 3!):

$$\vec{PQ} \times \vec{PR} = \langle -5, 21, -17\rangle$$

Step 3 — Area of triangle:

$$\text{Area} = \tfrac{1}{2}|\langle -5, 21, -17\rangle| = \tfrac{1}{2}\sqrt{25+441+289} = \tfrac{1}{2}\sqrt{755} \approx 13.74$$

The cross product of two edge vectors gives a **normal vector** to the plane — a vector pointing straight out of the surface. We'll use this constantly in Section 12.5.



Cross Products of the Standard Basis Vectors

The standard basis vectors $\vec{i}$, $\vec{j}$, $\vec{k}$ follow a **cyclic pattern**:

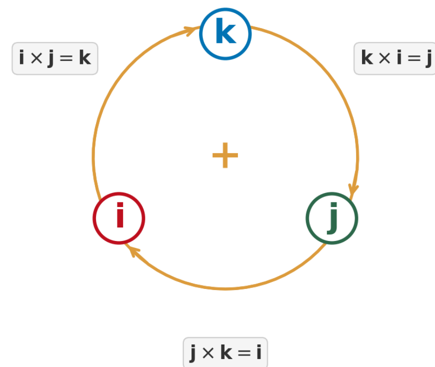
Forward (cyclic order $\vec{i} \to \vec{j} \to \vec{k} \to \vec{i}$):

$$\vec{i} \times \vec{j} = \vec{k}, \qquad \vec{j} \times \vec{k} = \vec{i}, \qquad \vec{k} \times \vec{i} = \vec{j}$$

Backward (reversed order = negative):

$$\vec{j} \times \vec{i} = -\vec{k}, \qquad \vec{k} \times \vec{j} = -\vec{i}, \qquad \vec{i} \times \vec{k} = -\vec{j}$$

Remember the cyclic pattern: $\vec{i} \to \vec{j} \to \vec{k} \to \vec{i} \to \cdots$ Going forward gives a positive result; going backward gives a negative.



Forward (clockwise) = positive
Reverse = negative

Homework

Section 12.4: 2, 3, 13, 16, 17, 19, 27, 29, 43, 44